MENOUFIA NATIONAL UNIVERSITY
Faculty of Computers & Artificial Intelligence

PROJECT REPORT

Sales Analysis Report

Reseller-Based B2B System

Data Analysis Course | Academic Year 2025–2026

■ PROJECT TEAM ■

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1. Project Overview

This report presents a comprehensive data analysis project focused on a company operating through a Reseller-Based Sales model (B2B). The company sells products to resellers (intermediaries), who in turn sell to end customers.

Dataset Description

- Sales Data – transactional records of all sales activities
- Products – product catalog with pricing and categorization
- Date – temporal data enabling time-series analysis
- Resellers – information about distribution partners

Note: The dataset does not include end-customer data directly linked to the sales table. All analysis focuses on reseller performance.

2. Project Objectives

- Analyze overall sales performance across time periods
- Understand revenue and profit sources within the B2B model
- Identify top-performing products by sales volume and revenue
- Evaluate reseller performance and rank key business partners
- Support strategic decision-making with data-driven insights

3. Key Analytical Questions

The analysis is structured around five core business questions:

#	Key Question	Business Purpose
Q1	Total Sales & Profits	Measure business size
Q2	Sales Change Over Time	Identify trends
Q3	Top-Selling Products	Identify strong products
Q4	Top Resellers	Identify key partners
Q5	Discount Impact on Profits	Measure impact of offers

4. Analytical Methodology

4.1 Data Preparation

- Data cleaning and handling of missing values
- Date formatting and calendar table creation
- Relationship modeling between Sales, Products, Resellers, and Date tables

- Calculated columns and measures using DAX

4.2 Analysis Approach

- Descriptive analysis – summarizing historical sales performance
- Time-series analysis – identifying seasonal and annual trends
- Comparative analysis – ranking products and resellers by performance
- Profitability analysis – evaluating the impact of discounts on profit

5. Data Visualizations & Dashboard Analysis

Dashboard 1: Sales Performance Overview

Figure 1 — Key Performance Indicators

Four top-level KPIs anchor the dashboard: Total Sales of $9.78M, Total Cost of $9.39M, 152 total orders, and 13K total units — establishing the business scale and revealing a thin profit buffer that requires careful cost management.

Chart 1: Total Sales by Day

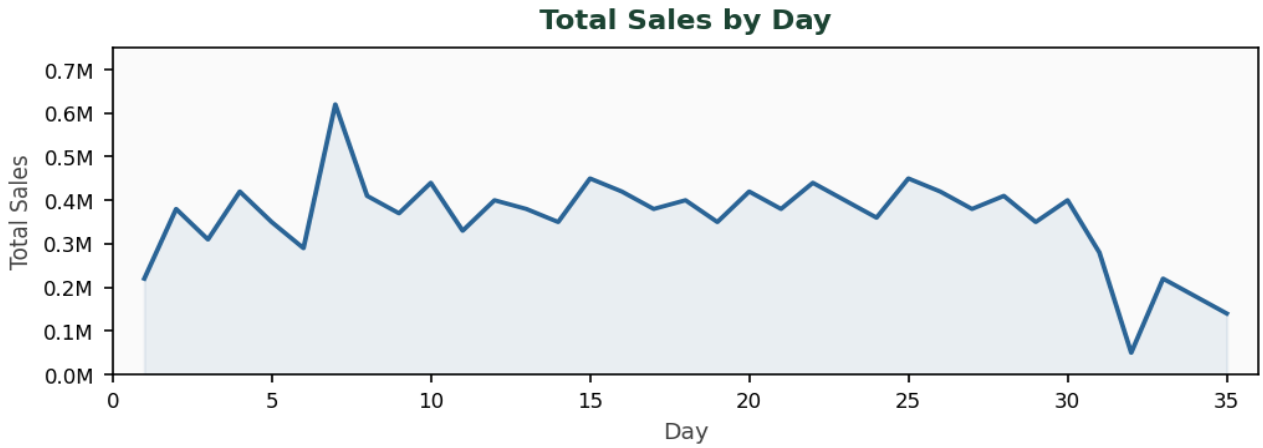

Figure 2 — Total Sales by Day (Line Chart)

This line chart tracks daily sales over one month. Values range from near $0.0M on the lowest days to a peak of ~$0.6M around Day 7, with most days between $0.2M–$0.4M — characteristic of B2B batch-ordering cycles. The sharp drop near Day 32 reflects end-of-period reduced activity.

Dashboard Analysis Questions

Q1. Describe the sales trend. On which day did the highest peak occur and what was its approximate value?

Answer: Sales are highly volatile. The highest peak occurs around Day 7, exceeding ~$0.6M. Most other days fluctuate between $0.2M and $0.4M, consistent with B2B reseller batch-ordering behavior.

Q2. What does the sharp drop near Day 32 suggest?

Answer: It signals the end of a billing or reporting cycle during which resellers pause new orders — a common pattern in B2B systems.

Q3. What action would you recommend to address the sales volatility?

Answer: Investigate peak-day drivers (e.g., promotions or reorder cycles) and implement reseller incentives to smooth monthly demand and reduce operational strain.

Chart 2: Sum of Profit by Product

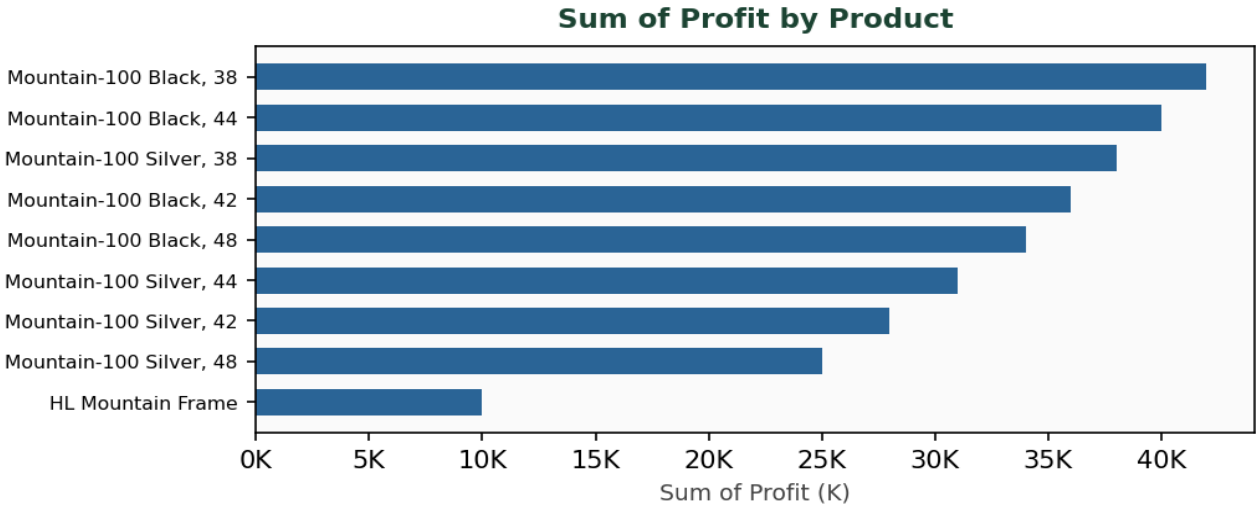

Figure 3 — Sum of Profit by Product (Horizontal Bar Chart)

This chart ranks products by total profit. Mountain-100 Black (38) leads with profit exceeding 40K, followed by other Mountain-100 variants. The HL Mountain Frame sits at the bottom (~10K), indicating it may require a pricing strategy review.

Dashboard Analysis Questions

Q4. Which product generated the highest total profit? Name at least three products in the top five.

Answer: Mountain-100 Black, 38 leads (>40K). Other top-five products include Mountain-100 Black (44), Mountain-100 Silver (38), and Mountain-100 Black (42).

Q5. What does the dominance of the Mountain-100 series tell us?

Answer: The Mountain-100 line is the company's core value driver. Prioritising this series in inventory, marketing, and reseller promotions will maximise profitability.

Q6. What action would you recommend for the HL Mountain Frame?

Answer: Review its pricing strategy, consider bundling with higher-margin items, or evaluate whether to continue stocking it at current volumes.

Chart 3: Sum of Profit by Category

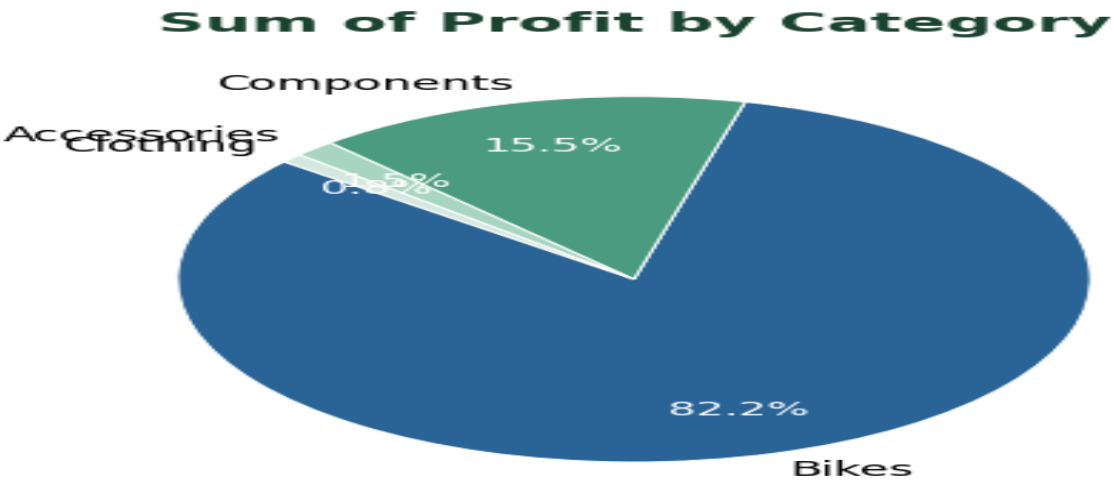

Figure 4 — Sum of Profit by Category (Pie Chart)

The Bikes category dominates with 82.2% of total profit ($322.3K). Components account for 15.5% ($60.78K), while Accessories and Clothing contribute a minimal share — highlighting a significant concentration risk.

Dashboard Analysis Questions

Q7. Which category dominates total profit and what percentage does it represent? What business risk does this concentration create?

Answer: Bikes account for 82.2% ($322.3K). This creates strategic risk — any demand shift or margin pressure in this segment would severely impact overall profitability.

Q8. Which category should management prioritise for growth?

Answer: Components (15.5%) is the most logical growth target as it already has a meaningful base. Expanding it through new product lines or reseller incentives diversifies revenue without requiring entirely new market entry.

Detailed Product Performance Table

The table below provides a granular view of Total Sales per product SKU, supporting inventory and supply chain planning decisions.

Product	Total Sales
Mountain-100 Black, 38	$766,938.13
Mountain-100 Black, 44	$750,844.83
Mountain-100 Black, 42	$705,443.11
Mountain-100 Silver, 38	$701,757.94
Mountain-100 Black, 48	$670,273.01
Mountain-100 Silver, 44	$660,419.50
Mountain-100 Silver, 42	$628,853.99
Mountain-100 Silver, 48	$542,638.40
Road-150 Red, 56	$461,596.83
Road-450 Red, 52	$377,036.21
TOTAL	**$9,782,608.34**

Table 1 — Detailed Product Performance

Dashboard Analysis Questions

Q9. Which product has the highest Total Sales? How does Mountain-100 Black (38) compare to Mountain-100 Silver (48)?

Answer: Mountain-100 Black, 38 leads with $766,938.13 — approximately $224K higher than Mountain-100 Silver, 48 ($542,638.40), confirming stronger commercial performance for the Black variant.

Q10. With 13K total quantity and 152 total orders, what is the average quantity per order and what does this reveal about reseller behavior?

Answer: Average = 13,000 ÷ 152 ≈ 85 units/order. This high average is consistent with bulk B2B purchasing — resellers place large batch orders rather than frequent small orders typical of retail consumers.

6. Key Findings & Insights

Sales Performance

- ◆ Peak sales days show 3× the volume of slower days, suggesting cyclical ordering patterns.
- ◆ The company maintains consistent year-round activity across its reseller network.

Product Insights

- ◆ The Mountain-100 series (Black and Silver) is the top revenue and profit driver.
- ◆ HL Mountain Frame products contribute lower margins and need a pricing review.

Profitability

- ◆ Total Sales ($9.78M) vs Total Cost ($9.39M) leaves a thin profit buffer.
- ◆ Bikes category generates 82.2% of profit, creating significant concentration risk.

7. Conclusion & Strategic Recommendations

This analysis delivers actionable insights into a B2B reseller-based business. The following strategic actions are recommended:

- ◆ **Cost Optimisation:** The narrow gap between total sales and cost requires renegotiating supplier pricing or reducing operational overhead to protect profitability.
- ◆ **Product Portfolio Focus:** The Mountain-100 series consistently outperforms all other lines. Prioritise inventory allocation and marketing toward this product line.
- ◆ **Category Diversification:** With Bikes at 82.2% of profit, growing Components and Accessories reduces strategic risk from over-reliance on a single category.
- ◆ **Reseller Performance Program:** Implement a tiered incentive program for resellers based on order volume and revenue contribution to elevate mid-performing partners.

Limitations: *End-customer data was unavailable, limiting consumer-level segmentation. Future work should incorporate predictive modelling to forecast sales trends and identify at-risk reseller relationships.*

8. Data Source

The dataset used in this project is the **AdventureWorks Sales Sample**, an official Microsoft Power BI sample dataset.

■ **Dataset Repository**	AdventureWorks Sales Sample — Microsoft GitHub

https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Fraw.githubuserco...

This report was prepared as part of the Data Analysis Course — Menoufia National University | Faculty of Computers & Artificial Intelligence | 2025–2026